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Impact Assessment of the Seeds for Needs Initiative in India

A Case Study of Bihar

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The **Development Impact Unit (DIU)** of Bioversity International is a highly-specialized research team that ensures the organization's research initiatives achieve the organization's strategic objectives of safeguarding agricultural and tree biodiversity, and diversifying diets, production systems, seeds and planting materials. The DIU connects, harmonizes and facilitate learning across the initiatives providing tools, methods and approaches for Impact Assessment (IA) analysis, knowledge and data sharing, science communication and gender-sensitive research.

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Executive Summary

This study has been conducted by Bioversity International's Development Impact Unit (DIU) upon commission from Bioversity's Board of Trustees. The goal is to assess the development outcomes and impacts for end-users of the Seeds for Needs (S4N) intervention in India. Seeds for Needs is a Bioversity International research initiative that promotes effective and cost-efficient ways to broaden farmers' seed knowledge and choices in the context of changing climatic conditions. In India, six projects falling under the S4N framework have been implemented between 2010 and 2017. Main donors include the Indian Council for Agricultural Research (ICAR), United Nations Environmental Program (UNEP), Food and Agriculture Organization (FAO) and the International Water Management Institute (IWMI), for a total investment of US\$8 million.

The S4N initiative's development goal is to increase households' capacity to adapt to the slow onset of climatic change by introducing and applying product and process innovation to agriculture - the major income, employment and food source of rural household livelihoods in the area under review. To achieve this, different types of training have been conducted to increase farmers' awareness of agricultural best practices and the role played by intra-species genetic diversity. The focus of S4N is on participatory seed selection, and newly developed rice and wheat varieties have been supplied for field testing using a "citizen scientist approach". The underlying assumption is that, whatever the environmental conditions, the more the agricultural system is diversified, the more likely it will be that some species will thrive under those conditions. Extensive literature, partially reported in the present study, shows that agricultural diversity strongly correlates with reduction in yield variance and disease suppression.

The evaluation is based on the results of a quantitative household survey, conducted between February and August 2018, on a stratified sample of 600 households in the state of Bihar. The household questionnaire is an adapted version of the Rural Household Multi-Indicator Survey (RHoMIS). The impact analysis employs a quasi-experimental, quantitative method design to understand the role played by the S4N intervention on the farm varietal diversification and capacity of target households to adapt to climate-induced agriculture shocks. In particular, the study aims to understand households' perceived agency and empowerment following the occurrence of shocks, and to establish a causal relationship between this perception, farm varietal diversity and yield fluctuations.

To identify and isolate a causal relationship between genetic diversity conservation and capacity to adapt and recover from the negative impacts of climate change on agriculture, the analysis blends the quantitative statistical technique of three-stage least squares multivariate regression (3SLS) and doubly robust estimation (DR). The former methodology is used to assess the extent of interdependency between the outcomes of (1) participation in project activities, (2) adoption of project varieties, and consequent (3) yield stabilization. The latter explores the impact that participation in the project has on households' capacity to recover from agriculture-related shocks, and examines the differences between treatment and control groups in terms of rate of recovery following shock occurrence.

Using different quantitative methods as complementary sources to paint a picture of the ways in which the Seeds for Needs Initiative has changed lives and farm landscapes, we can draw causal links. This causal story begins with project beneficiaries both adopting the disseminated varieties in 85% of cases and becoming generally more sensitive towards the role of intra-species genetic diversity in their farms. Despite the distress suffered by the agricultural sector in the study area, due to the frequency of climate-related shocks, greater genetic diversity in the target crop cultivation has led to lower yield variance scores. Improved stability of farm production has, in turn, led to participants feeling more empowered in responding to disturbances. Farmers perceived an increase in their capacity to recover from the occurrence of agriculture-related shocks since the intervention's implementation, as reflected

in their higher recovery rate scores. Moreover, this impact analysis has found that the S4N Initiative has had a beneficial effect on other household-level outcomes of interest, particularly a greater sense of social cohesion.

The study adopts the following structure: Section 1 contextualizes the Seeds for Needs Initiative and its rationale; Section 2 summarizes the main achievements, funding sources and activities of S4N implemented in India, as reported in the project's progress and final reports examined by the DIU. Section 3 explains the research methodology adopted, including the stratified sample developed, the questionnaire content, and the multivariate statistical approaches. Section 4 explores the results of project outcomes and impacts. Ultimately, conclusions are drawn in Section 5, followed by consideration on programme learning, design, and additional research.

1. Seeds for Needs Initiative Rationale

Seeds for Needs (S4N) is a Bioversity International research initiative designed to promote effective and cost-efficient ways to broaden farmers' seed knowledge and choice. S4N provides smallholder farmers with vital information on and better access to seeds of varieties that are the most adapted to their needs, contributing to lasting solutions for resilience and climate adaptation (Proietti and Ortolani, 2017). Currently, the Seeds for Needs framework encompasses 27 projects implemented in 14 countries in Africa, Asia and Central America.

While there are differences in the activities implemented in each location, these projects pursue common objectives and share similar approaches. Each one of them represents a building block in the implementation of the overall strategy, moving the intervention along the impact pathway, and providing learning and feedback for the improvement of the Initiative's overall design. Proietti and Ortolani (2017) summarize as follow the main goals and the activities planned:

Objective 1: Enhanced conservation and use of genetic resources for increased crop resilience and productivity

- Mapping climate change scenarios in and across different agro-ecological zones
- Analysis of the adaptive capacity of smallholder farmers
- Development of participatory and IT-based approaches for crops and varieties assessment and selection
- Identification and characterization of most preferred and best adapted varieties to diverse climatic scenarios
- Design of climate-smart agricultural practices and integrated strategies for conserving genetic resources.

Objective 2: Sustainable options for seed systems serving smallholder farmers

- Assessment of existing seed systems
- Development and testing of models for seed supply serving smallholder farmers
- Capacity development approaches for strengthening technical and managerial skills of community-based and small and medium enterprise (SME) seed business.

Objective 3: Creating an enabling environment for solutions at scale

- Evidence generation on the effectiveness of agrobiodiversity-based solutions to increase farmers' resilience to climate change and improve productivity
- Assessment and/or development of (new) mechanisms to support informed decision-making processes
- Identification of methods and tools for climate-resilient land use planning.

Major stakeholders range from scientific institutions to governmental organizations at different political levels, and also include development, non-governmental and community-based organizations. Partnership strategies are different in each country, responding to the local context and reflecting the relationships built by Bioversity International over the years. Partners have a common commitment to the Seeds for Needs approach and are key actors in driving progress in the impact pathway and scaling up at a local level.

The S4N Initiative's Scientific Basis

A multitude of studies covering a wide range of regions and crops report negative impacts of climate change on crop yields, and notice how risks for human populations are unevenly distributed, being generally greater for marginalized individuals in all countries regardless of their development level (IPCC, 2014). In regions where communities are highly dependent on agriculture for their livelihoods, and are therefore particularly sensitive to climatic variations, the development and deployment of stress-tolerant seed varieties is vitally important.

Nonetheless, the intrinsic and unique features of agriculture make agricultural research and development (R&D) particularly challenging, among which site specificity and obsolence are major obstacles (Pardey, 2010). Site specificity circumscribes the potential for spillovers and diffusion. While research applied to other disciplines and sectors usually exhibits significant capacity for the exploitation of economies of size, scale and scope, much agricultural technology is characterized by the specificity of agroecological conditions, limiting the relevant market. The fall of unit cost that comes with the size of R&D must thus be traded off against diseconomies of distance and adapting site-specific results (Pardey, 2010). Similarly, obsolence highlights how resources and practices need to evolve over time in response to environmental, climatic and biological changes. Enhancing “gene by environment” interaction is an intrinsic feature of agriculture, characterized by an ever-changing geographical and environmental footprint (Pardey, 2010).

The S4N framework design embeds a mechanism to potentially overcome these barriers. Project proposals and reports (See for example Mathur 2011, 2012, 2013, 2014 for the case of India), consultancy evaluations (Proietti and Ortolani, 2017), scientific publications (Mancini et al, 2017; Van Etten et al, 2016; Mitra et al, 2015) and institutional communications (Bioversity International, 2015) all highlight the potentiality of the participatory approach, where farmers’ are “put at the centre of research activities and are involved as citizen scientists”. Farmers are thus exposed to a range of seed varieties potentially suitable for the agroecological zone they need to cultivate, and offer their fields for on-farm experimental trials of these seeds.

Capacity building activities are a fundamental component of the S4N framework (Proietti and Ortolani, 2017). Thanks to the close collaboration with scientific staff, participating farmers receive the necessary capacity to test the varieties that best match their preferences and needs, and to identify and replace the obsolete, low performing varieties and unsustainable agricultural practices. At the same time, they are given the opportunity to exchange information and best practices with their existing network. (Mathur, 2011). The dissemination channel appears to be that of local seed systems, as increased climate-resilient agrobiodiversity solutions are also made available to producers not directly targeted by the project activities. The direct and indirect outcomes of these activities are expected to contribute to the livelihood stabilization of natural resources-dependent households because they maximize adopters’ capacity to adapt to climate change and, ultimately, enhance the local social welfare (Proietti and Ortolani, 2017; Singh, 2017; Mathur, 2011-2012).



Figure 1: Bioversity India staff member, S. Kumar, collecting information on seed performance from a S4N participant in Vaishali, India. Credits: Bioversity International, S. Kumar

2. Background on Seeds for Needs in India

The implementation of the Seeds for Needs (S4N) Initiative in India started in 2010 in the Vaishali district of Bihar, the site selected for implementation of climate-smart agriculture under the CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS). Since 2011, the Initiative has been extended to nine more states: Uttar Pradesh, Odisha, Madhya Pradesh, Chhattisgarh, Orissa, Punjab, Haryana, Jammu and Kashmir (Singh, 2017). Several projects contributed to the S4N Initiative (Table 1) and find common ground in their long-term goal to **promote conservation and use of plant genetic resources, both as a means to reduce farmers' vulnerability to climate change and as an objective *per se***. Most of the S4N projects focused exclusively on the major crops in the country, i.e. rice and wheat, while a few have later projects have also supported the distribution of other crops including sesame, soybean, mustard, sorghum, and red, black and green gram. However, because of the prominent attention given to rice and wheat, this specific evaluation focuses exclusively on these two crops.

Table 1: Projects mapped under the S4N framework in India

Project Title	Donors	Period	Amount
Initial establishment of a farmer-based experimentation network in the Indo-Gangetic Plains (IGP) region: Pilot project for on-farm participatory research - Phase I (Wheat)	IWMI	2010/2011	\$50.000
On-farm participatory network for climate change adaptation and visualization in the Indo-Gangetic Plains (IGP) - Phase II (Rice)	IWMI	2011/2012	\$50.000
Using rice genetic diversity to support farmers' adaptation to climate change for sustainable production and improved livelihoods in India	FAO	2012/2014	\$296.000
Seeds for Life – Action with farmers in Uttar Pradesh – IGP region to enhance Food Security in the context of Climate Change	FAO	2012/2014	\$299.000
Use and conservation of agrobiodiversity for increased agricultural sustainability, smallholder wellbeing and resilience to climate change in India	ICAR	2013/Ongoing	\$5.000.000
Mainstreaming agrobiodiversity conservation and utilization in agricultural sector to ensure ecosystem services and reduce vulnerability - PPG stage	UNEP/ GEF	2013/2015; renewed in 2017	\$3.064.000
Total Investment		2010/Ongoing	\$8.800.000

The Initiative's main actors comprise a variety of national research institutes and smallholder farmers, the latter being the intended principal next users. Bioversity International has played a communication and networking role and has acted as a bridge to bring together different stakeholders. A list of S4N INITIATIVE partners for India can be found in **Annex I**. Figure 2 illustrates the reconstructed theory of change for the S4N projects. A list of project documentation reviewed for its development is available in **Annex II**. The general trajectory of the Initiative's scaling-out for research institutes and next users are coloured in blue and green, respectively, to illustrate the process of initial inputs and the consequent outputs that have led to the outcomes. Issues that the Initiative seeks to resolve are coloured red, while detailed Initiative outputs are in purple. Desired impact of the initiative is located to the far right, while consistency with the Initiative's strategic objectives, as mentioned in the previous section, are highlighted in yellow. The grey boxes indicate the outputs, outcomes and impacts the study seeks to assess.

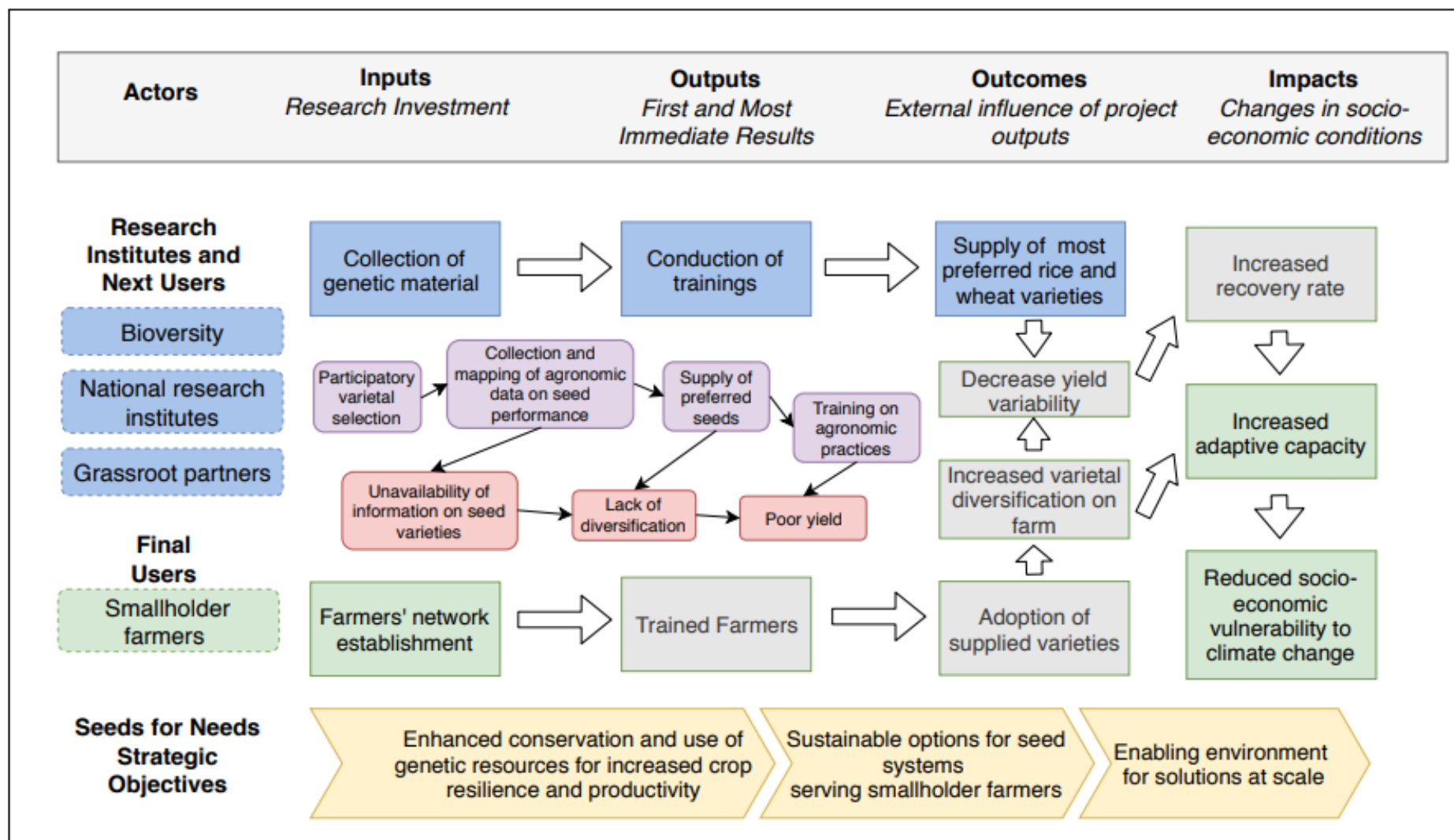


Figure 2: Theory of Change for Seeds for Needs India

The initial research investments included promotion and networking activities aimed at building partnerships with national research institutes and grass roots organizations with strong ties to local farming communities. The former supported the identification of suitable rice and wheat varieties, which had to be made readily available for field-testing. The latter contributed to the kick-off of field trials by establishing an initial network of farmers willing to participate in the S4N projects. A list of tested varieties is provided in **Annex III**.



Figure 3: Farmers participating in S4N in Bihar, receiving seeds for the field trials. Credit: Bioversity International/ S. Kumar

The outputs of this collaboration, defined in the Theory of Change as “first and most immediate results”, are a number of trainings conducted by researchers to farmers, with a particular attention to the inclusion of women. Activities focused mainly on “participatory varietal selection”, such as crowdsourcing and participatory trials¹, where farmers are supplied with random varieties of rice and wheat selected from a pool of potentially suitable seeds depending on their agroecological zone of cultivation. Farmers grew these varieties in small plots and autonomously chose the best performing varieties, providing feedback on preferred traits to scientists regularly visiting their fields for the collection of agronomic data. In addition, the local partner Krishi Vigyan Kendras (KVK) trained participants on seed conservation and reproduction practices, while also supporting the establishment and maintenance of 11 community seed banks, 7 of them directly operated by community members from the villages. Lastly, other important activities include farmers’ field days, creation of farmers’ field schools and exchange visits between sites to increase the farmers’ exchange of information on seed cultivation and agronomic practices and to promote collaborations among different communities while at the same time increasing the existing network of participants through word of mouth. This process allowed the supply and testing of 103 varieties of wheat and 128 varieties of rice. Georeferenced data on their performance has been collected and mapped. Approximately, 2500 farmers have been trained (Figure 4) (Bioversity Int. India, factsheet n.4).

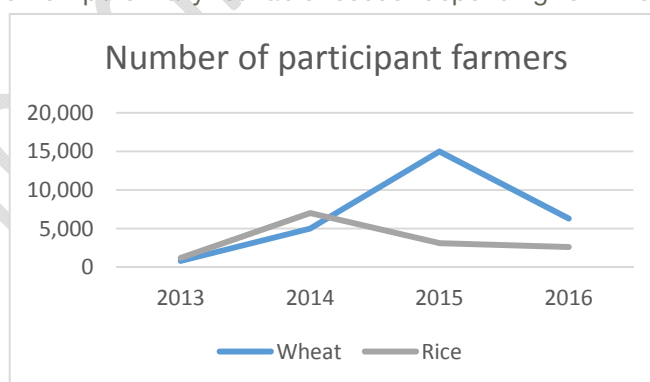


Figure 4: Number of S4N participating farmers in India

The next sections of the report follow the structure of the Theory of Change. Section 4 provides a detailed explanation of the research methods adopted and the variables of choice for the evaluation of the above-mentioned outputs, outcomes and impacts, which, for the econometric purpose of the study, are referred to as *outcome variables*. Thereafter, results are presented and discussed in the same order.

¹ Crowdsourcing trials (CS) and Participatory varietal selection (PVS) trials are different in nature yet pursue the same goal. PVS are more technical in nature and aim at providing quantitative data, as farmers test 10 to 20 varieties each time. The best performing varieties are then selected for CS, where 3 varieties are tested each time, allowing for a qualitative assessment of their performance.

3. Study Design and Methodology

This impact analysis was commissioned to understand the impact the S4N intervention had on the livelihood adaptation, varietal diversification and capacity to absorb climate shocks of target households. The study employs a quasi-experimental, quantitative methods design, which blends the statistical technique of three-stage least squares multivariate regression (3SLS) and doubly robust estimation (DR) to rigorously and extensively evaluate whether, and to what degree, the project had an impact on beneficiary households.

3.1 Study design

The framework of reference for the methodological design is that of vulnerability studies. The concepts of vulnerability, resilience and adaptive capacity are relevant in both the social and the life sciences. They are widely used in sustainable development research to study Socio-Ecological Systems (SES) (Figure 5), pointing out the societal (human) and ecological (biophysical) characteristics of a system and their mutual interactions (Gallopin, 2006). While measurements of vulnerability are subject to significant differences in the literature (see for example Han et al, 2009; Ahsan and Warner, 2014; Badmos et al, 2018), scholars mostly agree on disentangling the concept into three components: exposure, sensitivity and adaptive capacity (Otto et al, 2017; Adger, 2006). Exposure refers to the general duration and/or extent to which the system is subject to a perturbation. Sensitivity is defined as the degree to which the system is modified by a disturbance, positively or negatively. Adaptive capacity is the system's ability to adjust to disturbances, cope with the consequences, mitigate damages and take advantage of opportunities. A system is adaptive when it is able to evolve and accommodate change to "expand the range of variability" it can cope with, and is often understood as a manifestation of resilience (Adger, 2006).

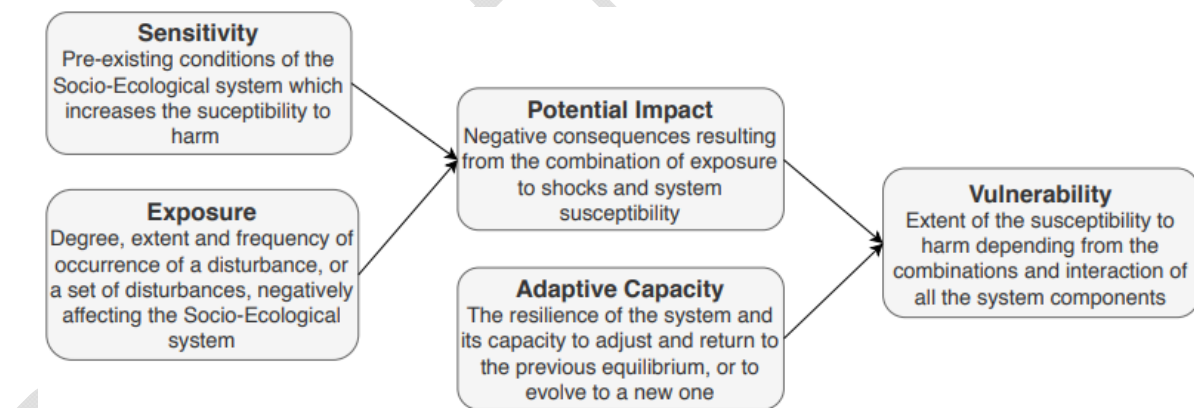


Figure 5: Theoretical framework of the vulnerability of Socio-Ecological Systems

From a social and human perspective, adaptive and resilient households are able to anticipate and resist future stresses, and regenerate themselves according to their motivations and capabilities. Social resilience refers to the adjustment undertaken to "moderate harm and exploit beneficial opportunities" in response to climatic stimuli (IPCC, 2007). However, it is recognized that individual risks are not equally distributed within society, mostly because of inequalities and conditions that exist prior to the occurrence of the destabilizing event, and people's everyday life habits, practices and characteristics.

From an agroecological perspective the diversity of farm production functions as a buffer against environmental fluctuations because different species respond differently to change, thus leading to a more predictable and potentially stable agroecological system (Lin, 2011). The underlying assumption is that whatever the environmental conditions are or will be, the more the agricultural system is diversified, the more likely it will be that some species are able to thrive under those conditions (Heal, 2000). Agricultural diversity strongly correlates with reduction in yield variance (Smale et al, 1998; Widawsky & Rozelle, 1998; Di Falco and Chavas, 2009) and disease suppression (Heisey et al, 1997; Zhu et al, 2000). Evidence of yield productivity increase is instead mixed, highly complex and context specific (Smale et al, 1998; Widawsky & Rozelle, 1998; Nankya et al., 2017). The relationship between diversity and risk reduction is known in the literature as the “Insurance hypothesis”, an intuitive idea that increasing species richness insures ecosystems against a decline in their functioning because of a decline in the variability of their processes (Yachi & Loreau, 1998). The stabilization of agricultural production has major effects on the agriculture-dependent households, especially through the consequent stability of income (Di Falco & Perrings, 2003).

The linkages between the diversity of agroecosystems to their function capacity and resilience, and the consequent livelihood benefits for households dependent on agriculture, are the very basis of the S4N Initiative framework. In applying the vulnerability framework to our case study in Bihar, we consider both the degree of exposure to perturbations affecting the agricultural sector, and the levels of sensitivity in terms of demographic characteristics and viable livelihood strategies outside of agriculture. However, because these components are not within the sphere of action of Seeds for Needs, the focus of the impact analysis is on adaptive capacity and resilience. In particular, the study aims to understand households’ perceived agency and empowerment following climate-related shocks, and to establish a causal relationship between this perception, varietal diversification on farm, and yield fluctuations.

3.2 Sampling framework and data collection methodology

A common problem that arises when analyzing the effect of a specific project ex-post, is the estimation of a counterfactual scenario in the lack of baseline data. A widespread practice is to make a comparison between units that were subject to the project and those that were not. Randomly placing individuals within the treatment and control groups ensures that on average any differences can be attributed to the project. One of the issues inevitably arising is bias due to non-random selection. Seeds for Needs implementing agencies targeted specific communities that were seen to be particularly vulnerable, and where there was potential for the project activities to have a positive impact. However, participants themselves determine whether or not to join the project, making it difficult to establish a causal relationship between the observed outcomes and impact and the participation in the project (Gotor et al., 2017).

To address the problem of selection bias, we leveraged key contextual factors. Firstly, logistical constraints meant that only a portion of households that would have been eligible for Seeds for Needs training actually were reached by the project. Yet, other households within the regions faced similar risks, and there were other communities with equally vulnerable groups where the project activities could have been implemented. This allowed us to assume that people living in communities not included in any of the projects would provide a reasonable counterfactual. Furthermore, because one of the peculiarity of Seeds for Needs was to exploit existing farmers’ social networks as a channel for the reproduction and dissemination of seed varieties distributed under the project, we opted for a stratified sample that would enable the identification of

spillovers. Therefore, people living in the same communities reached by the project who decided not to take part in it are also considered a control group. Our analysis is based upon 600 households, and the sample is stratified as follows (Table 2):

- **Treatment (T):** 300 project participants – intervention group who voluntarily expressed their willingness to participate;
- **Control 1 (C1):** 150 non-participants from the same villages of participants – comparison group 1;
- **Control 2 (C2):** 150 non-participants from from different villages, where the project was never implemented.

Table 2: Study Sampling Framework

Treatment			Control 1			Control 2		
Districts	Villages	Sample	Districts	Villages	Sample	Districts	Villages	Sample
Saran	Bhagwanpur	3	Saran	Bhagwanpur	15	Saran	Dharmagt Tola	19
Saran	Sultanpur	10	Saran	Sultanpur	24	Saran	Khanpur	19
Saran	Sabalpur	8	Saran	Sabalpur	13	Saran	Sembhopatti	20
Saran	Rampur Jaitti	18	Saran	Rampur Jaitti	21	Vaishali	Vishanpura	9
Vaishali	Bhathadasi	57	Vaishali	Bhathadasi	28	Vaishali	Fatehpur Chauthai	18
Vaishali	Kariyo	10	Vaishali	Kariyo	3	Vaishali	Kutubpur	23
Vaishali	Mukundpur	31	Vaishali	Mukundpur	2	Vaishali	Mirpur patadh	5
Vaishali	Panapur	4	Vaishali	Panapur	1	Samastipur	Dhobgama	20
Vaishali	Rajapakar	77	Vaishali	Rajapakar	10	Samastipur	Narayanpur	17
Samastipur	Harpur	32	Samastipur	Harpur	16			
Samastipur	Madapur	14	Samastipur	Madapur	5			
Samastipur	Mahamada	36	Samastipur	Mahamada	12			
Total		300	Total		150	Total		150

The household questionnaire was administrated between February and August 2018 in three districts of Bihar: Saran, Vaishali, and Samastipur (See the GIS map in Figure 6). The data collection team consisted of three enumerators. Enumerators attended a four-day training and field-testing series. One enumerator was designated team leader and was responsible for cross-checking all household data at the end of each day. The enumerators used electronic tablets to record the data using the Open Data Kit (ODK) platform. All data was uploaded to a server at the end of each day after being checked by the team leader.

The household questionnaire is composed of 17 sections, of which 3 are specifically devoted to impact evaluation and focus on: (1) the participation in project activities, (2) target crop cultivation, and (3) exposure to shocks and recovery capacity. The remaining sections are adapted from the Rural Household Multi Indicator Survey (RHoMIS) following enumerators' feedback during the training. RHoMIS is a household survey tool designed to rapidly characterize a series of standardized indicators across the spectrum of agricultural production and market integration, nutrition, food security, poverty and GHG emissions. (Hammond, 2017). A detailed description of questionnaire sections can be found in **Annex IV**.

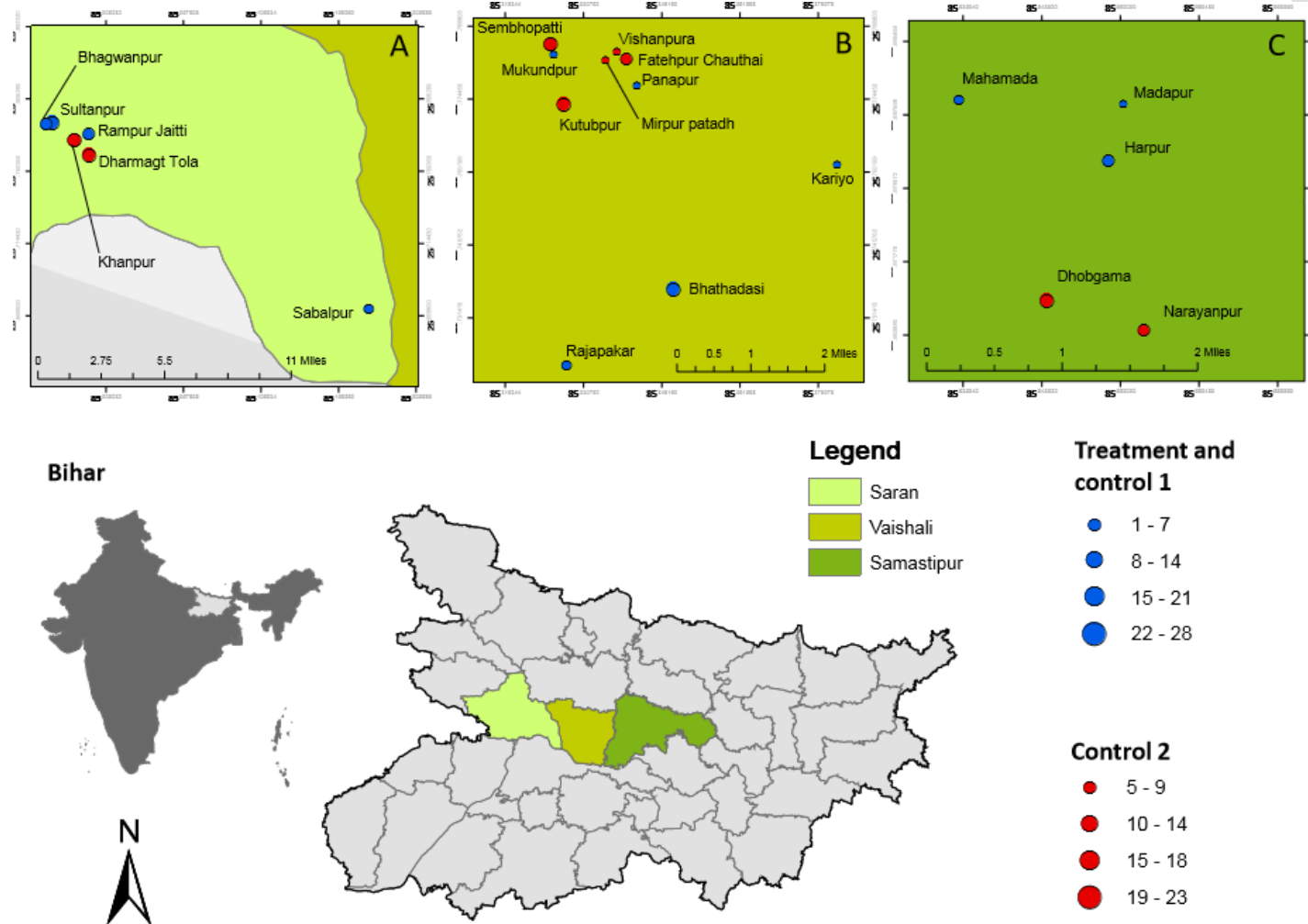


Figure 6: Map of selected villages – Credit: Bioversity International/G. Gullotta.

3.3 Quantitative Methods

- The analysis combines two quantitative methods of analysis in order to achieve a more comprehensive level of understanding of the project impacts. Firstly, a three-stage least squares regression (3SLS) is run to assess the consequentiality of relationship between project outputs and outcomes. Secondly, a doubly robust estimator of causal effects (DR) is performed to model the relations between confounders and the outcomes within each group. As indicator of project output **participation to project activities** was used: households were indeed invited to participate in four different trainings related to (1) participatory varietal selection, (2) seed conservation and reproduction, (3) agronomic practices and (4) events for the exchange of information with other farmers. For the purposes of the analysis, a dummy is considered if households participated to at least one training.

3.3.1 Definition of outcome variables

The first step is the identification of outcome measures², which are case-specific and consistent with the project aims. For the purpose of this study, the following outcomes are relevant:

- **Adoption of project varieties:** Considering the number of varieties the farmers were exposed to, a variable is created to measure the extent of adoption, i.e. the number of varieties disseminated through S4N that farmers were cultivating at the time of the survey and that they defined as very important for their production.
- **Farm Varietal Diversification:** As farmers were exposed to a number of rice and wheat varieties, and at the same time encouraged to personally evaluate the potential benefits of diversifying their farm production, the Simpson Index (SI) is used. It is the most suitable index for measuring diversification of crops in a particular geographical region and is calculated by:

$$\text{Simpson Index (SI)} = 1 - \sum_{j=1}^J P_j^2$$

where, $P_j = A_j / \sum A_j$ is the proportion of the j -th activity in acreage. If SI is near zero, it indicates that the household is near to the specialization of growing a particular crop variety and, if it is close to one, then the household is fully diversified in terms of varieties.

- **Yield Change:** The increase in yield stabilization is the channel through which the intended impact in terms of recovery capacity is achieved. Yield change is measured by:

$$\Delta Y = \frac{Y_t - Y_{t-1}}{Y_{t-1}}$$

- **Household Recovery Rate:** The intervention impact is measured in terms of increased capacity to respond and recover from the occurrence of shocks affecting agricultural production. This information is collected in the survey with a self-reported questionnaire covering the extent of exposure to agricultural-related shocks occurred in the timeframe 2015-2017, namely: (1) increase in food prices, (2) increase in

² For the purposes of the econometric analysis, we define outcome measures as all the variables of interest, without making the distinction of output, outcome and Impact as per the Theory of Change. However, results in the next section will be reported following the ToC order.

the prices of inputs, (3) decrease in sale prices of rice and wheat, (4) loss of rice and wheat harvest because of pest and disease, and (5) because of climatic stressors. For each shock, the perceived recovery rate was collected using a Likert scale-based set of answers. An index ranging from 0 to 100 is constructed to aggregate the variables under consideration using the formula:

$$\text{Recovery Index (R)} = 50 * \frac{L_m + 1}{5} * \left(2 - \frac{F_m}{4}\right)$$

Where L_m is the recovery level from the m -th shock and F_m indicates the frequency of shock occurrence, the algorithm was developed to award a higher score to households who suffered high frequency of shock occurrence but declared to have recovered completely. Similarly, also households that were not exposed at all receive a higher score. On the contrary, households highly exposed but with little perceived recovery score lower.

- Perception of the **trend of agricultural productivity** in the past year, measured with a Likert scale, to double check the validity of the Recovery Rate index specifically for agricultural productivity;
- **Social Cohesion:** defined as the willingness of households to cooperate with each other in order to survive and prosper. This information is obtained by aggregating a set of questions related to the level of trust in the community, the willingness to help each other, and the important gift received by family members or friends. It is obtained with the formula:

$$\text{Social Cohesion (SC)} = \sum G + T + C + L + B$$

where G is the number of important gift received; T is a dummy that indicates if the respondent thinks people can generally be trusted; C indicates the type of contribution that the respondent would be willing to offer to other members of the community; L is a subjective measure on the level of trust and cooperation within the community, and B indicates if the respondent would be willing to engage in business activities with the other villagers. Final score ranges from 0 to 16, with 16 indicating the maximum level of cooperation and trust. While this score is not explicitly intended as an impact of the project, the research team believe its increase could represent an unintended consequence of farmers' field days and exchange visits, which were aimed at increasing the network of farmers.

3.3.2 The Three-Stages Least Squares Regression Model

The first multivariate analysis consists in the provision of a clear reference to the dynamic process explained in the Theory of Change (ToC), meaning the consequentiality of the steps linking the livelihood benefits (i.e. the impacts on the capacity to recover) to the participation in the project training.

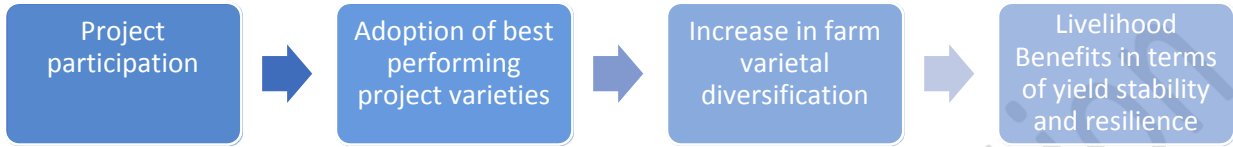


Figure 7: Zoom on the output, outcomes and impacts of interest for the study

To establish an explicit causal relationship, it is necessary to evaluate how participation itself affects the adoption rate of project varieties, how the adoption eventually leads to increased varietal diversification, and how such diversification influences the yield change and the household rate of recovery. Therefore, the three-stage least squares (3SLS) regression model is performed. 3SLS is a method of estimation originating in a paper by Arnold Zellner and Henri Theil (1962), which combines system equation, sometimes known as seemingly unrelated regression (SUR), with two-stage least squares estimation, and estimates all coefficients simultaneously. It can be defined as a form of instrumental variables estimation that permits correlations of the unobserved disturbances across several equations. The 3SLS is thus used to capture the dynamic linkages among outputs (project participation) and outcomes (adoption, varietal diversification and yield variance), and it is formulated for the i -th household as follows:

- (1) $Adoption_i = x_i' \theta + \gamma Participation_i + v_i$
- (2) $Varietal\ Diversification_i = x_i' \alpha + \beta Adoption_i + u_i$
- (3) $Yield\ Change_i = x_i' \omega + \delta Varietal\ Diversification_i + e_i$

Where *Adoption*, *Varietal Diversification* and *Yield Change* are the three outcome variables; x_i is a vector of confounding factors; θ , α and ω are the parameter vectors of the equations' system that measure the effects of the exogenous variables on the outcome variables; while v_i , u_i and e_i are the error components. Finally, it is the estimation of the parameters γ , β and δ that allows us to test the three hypotheses simultaneously. Indeed, through the estimation of the parameter γ , the model measures whether participation in the project affects the adoption of the varieties disseminated through Seeds for Needs project (Equation 1). The β parameter quantifies the impact of the project on the varietal diversity of the i -th household (Equation 2), while the parameter δ measures the effects of this diversity on the variation of the yield (Equation 3).

3.3.3 Doubly Robust Estimator of Causal Effect

Once the causal relationship among project outputs and outcomes is established, a further multivariate analysis is run to estimate the average causal effect of the treatment (participation in the project trainings) on the following outcome variables: (1) varietal diversification, (2) yield change, (3) recovery rate, (4) perceived agricultural trend, and (5) social cohesion. For this purpose, the analysis relies on the doubly robust estimation to quantify the differences between treatment and control groups. This technique combines two different approaches to estimate the causal effect of an exposure on an outcome: a form of outcome regression and a model for the exposure (Funk et al, 2011). This ensures the robustness of the results because possible forms of misspecification of the model due to selection bias and confounding effects are controlled (Emsley et al, 2008; Wanjala and Muradian, 2013; Caracciolo and Furno, 2017).

$$DR = \frac{1}{I} \sum_{i=1}^I \frac{W_i Y_i - (W_i - \widehat{p}(X_i)) \widehat{Y}_{i1}}{\widehat{p}(X_i)} - \frac{1}{I} \sum_{i=1}^I \frac{(1 - W_i) Y_i + (W_i - \widehat{p}(X_i)) \widehat{Y}_{i0}}{1 - \widehat{p}(X_i)}$$

Y_i is the *observed* outcome, $\widehat{Y}_{i1}$ and $\widehat{Y}_{i0}$ are the *predicted* outcomes for the i -th household under the two exposure conditions: treated ($W_i=1$) and untreated ($W_i=0$); and $\widehat{p}(X_i)$ is the *predicted* probability of being exposed conditioned to a set of observed characteristics of the i -th household not affected by the project.

4. Results and Discussion

This section presents a comparison of the households interviewed in the project and comparison communities, in terms of various outcome measures relating to the Initiative under review. The findings are organized around the three main steps of the Initiative's Theory of Change (ToC) in terms of output, outcomes and impacts. It is important to stress that the results presented in this section are average results across the intervention and comparison groups. Where the results benefit from visual representations, charts and graphs illustrate the findings to further support the analysis. It is important to bear in mind that the statistical procedures used are based only on 'observable' characteristics. If there are any 'unobserved' pre-existing differences between the project participants and comparison respondents – such as individuals' attitudes or motivation, differences in local leadership, environmental or other contextual conditions – then these may affect the estimates of outcome variables. This possibility cannot be excluded and must be borne in mind when interpreting the results, because project participants were, to some extent, a self-selected group (Gotor et al., 2017).

4.1 Participation in S4N activities

Figure 8 reports the attendance rate at each training. The training sessions with the highest attendance rate are those on participatory varietal selection and events for information exchange, such as farmers' field days. The attendance rate is 88% and 97%, respectively. Farmers' field days have the highest attendance in terms of number of members of the same household who joined the activities, two on average. Trainings on seeds conservation and reproduction practices were attended by 55% of respondents, while those on agronomic practices by 45%.

Among participants, the inclusion of women is 38% on average for all the trainings, with the exception of agronomic practices where their participation is slightly lower (33%). Among all training, the average proportion of women attendees is equal to 49% when participating with their partner or other family members, while only 6% when attending alone.

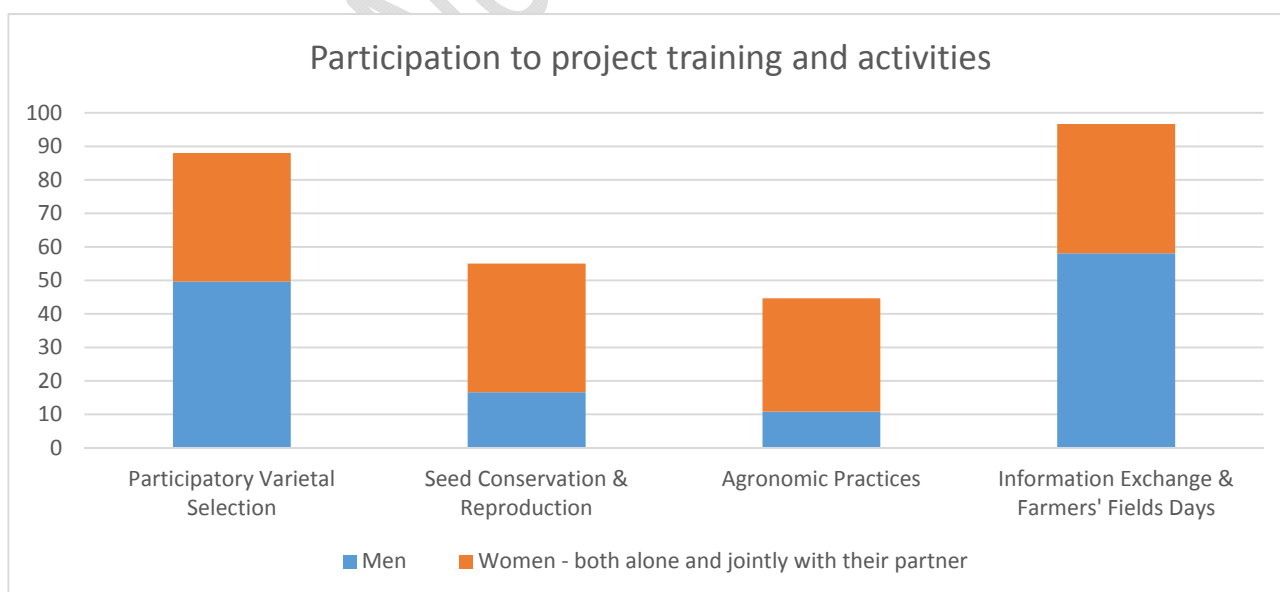


Figure 8: Gender-disaggregated participation in training

4.2 Adoption of Project Varieties

In order to understand if respondents actually appreciated and cultivated the varieties previously tested, respondents were asked a more detailed question on the ‘Most important rice and wheat varieties in terms of food and income’. Therefore, the study’s definition of adopter is that of “any household that mentions at least one Bioversity- or KVK-released variety among those most important for their livelihood”. The name of the released varieties was cross-checked with the seed source in order to understand if the seed was either provided by the S4N project or gained indirectly through the local market or farmer-to-farmer exchange – representing in this latter case a process of dissemination through spillover.

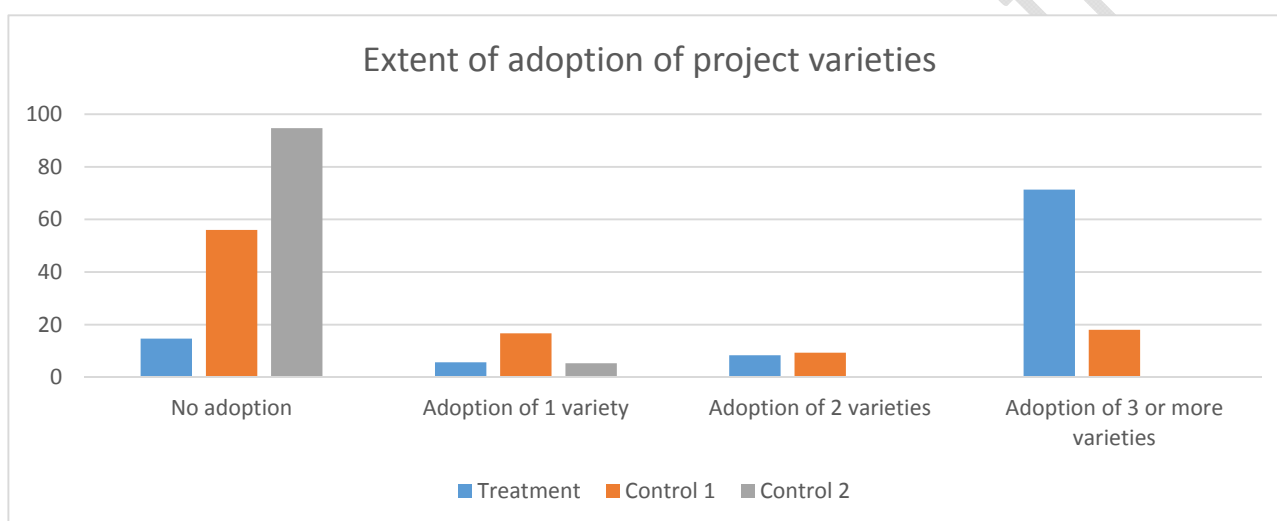


Figure 9: Extent of adoption of project varieties

Figure 9 shows the adoption rate of the target crops. Results are expressed in percentage for each number of adopted varieties, ranging from none to three or more. For both crops, about 15% of non-adopters belong to the treatment sample. These households have thus attended project activities, yet have decided not to keep any of the varieties proposed to them for future cultivation³. This percentage grows to 56% and 95% for non-beneficiaries from the same villages (Control 1) and non-beneficiaries from other villages (Control 2), respectively. Among the adopters of one variety it is possible to isolate the spillover effect: 17% of respondents from Control 1 and 5% from Control 2. An additional spillover emerges among the adopters of two varieties (9% from Control 1) and three varieties (18% from Control 1). Most of the treatment group, equal to 71%, have adopted three or more varieties.

85% of participants continue to cultivate the varieties they tested in the project. Most of them (71%) cultivate up to three or more of the project’s seeds. While the spillover effect is moderate for communities not reached by the S4N intervention (5%), it increases considerably when the geographical distance is reduced (18%).

³ While the present study does not account for an in-depth analysis of the reasons for discarding the seeds, it is important to mention that limited data has been collected on this matter. The reason most frequently reported by the households for discarding seeds is “seed accidental loss” and “seed not timely available”.

Focus on Rice

Adopters reported that the grain size (94%), yield capacity (93%) and cooking time (91%) are the best characteristics of project varieties that have led them to adoption. These are followed by market-related characteristics: the sale price of the harvest (66%) and demand (63%) (Figure 10). The most appreciated rice varieties among those released by the project intervention are Gontrabidhan (35% of adoption rate), Rajendrabhagwati (34%) and Sugandh 5 (28%) (Figure 11).

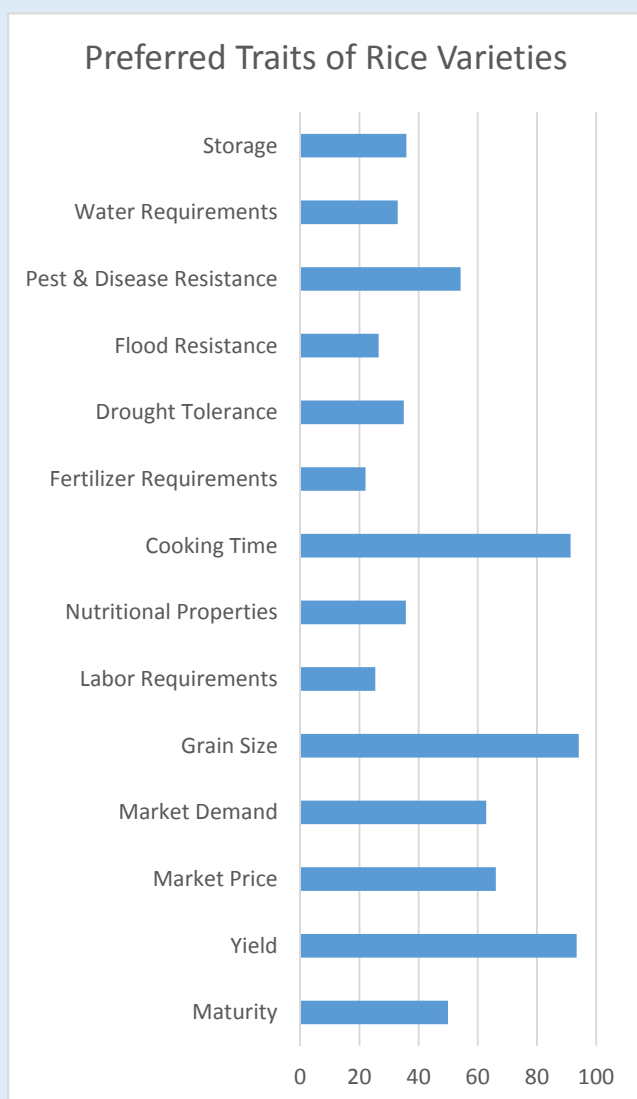


Figure 10: Rice varieties - preferred traits

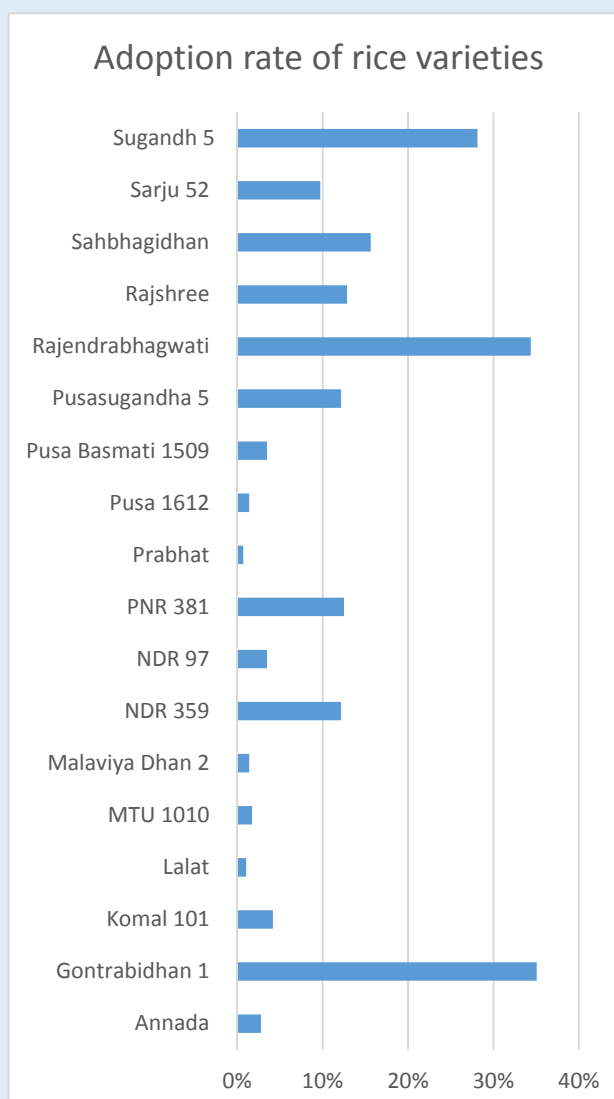


Figure 11: Rice varieties - extent of adoption

Focus on Wheat

As for rice, grain size (97%), yield capacity (95%) and cooking time (87%) are the best characteristics of wheat varieties that have led participants to adoption, followed by market demand for the harvest (70%), and the resistance to pests (75%) (Figure 12). The most appreciated wheat varieties are HD2967 (45% of adoption rate), HD2733 (39%), and PBW343 (35%) (Figure 13).

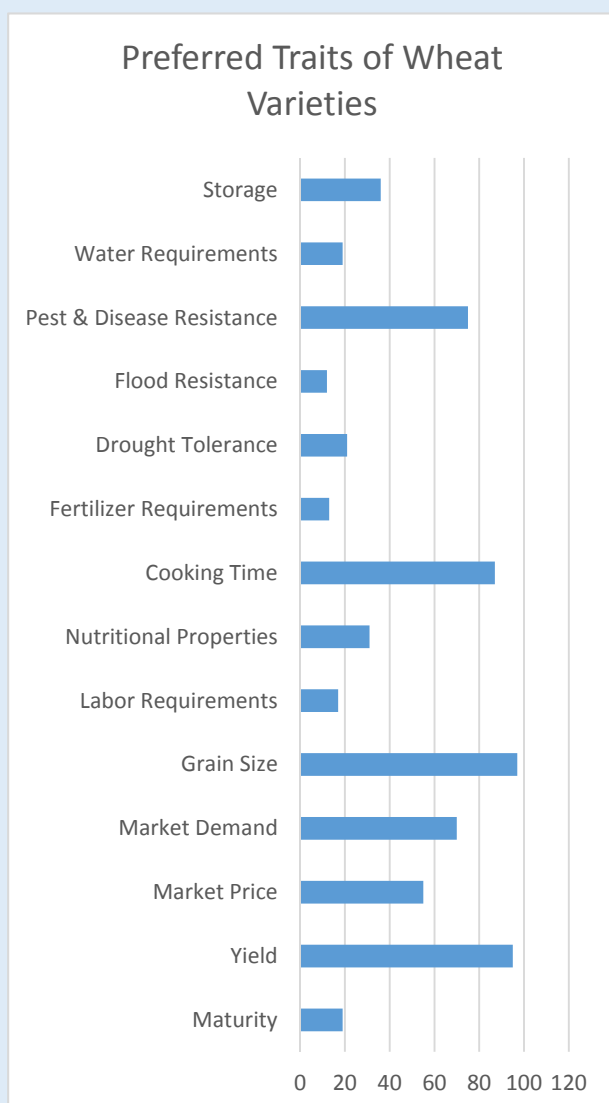


Figure 13: Wheat varieties - preferred traits

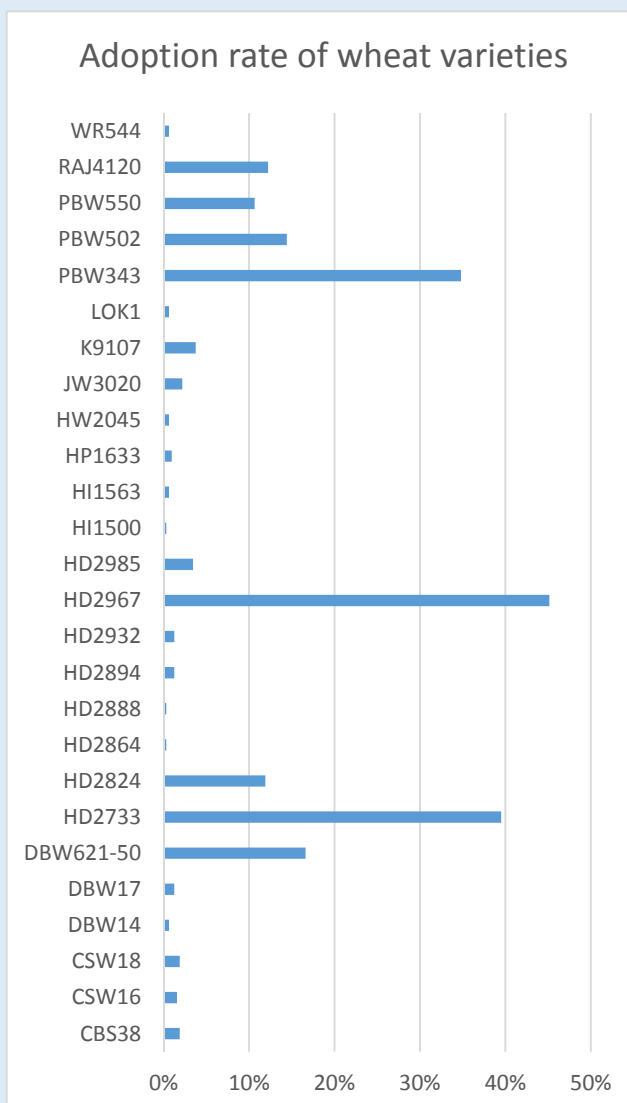


Figure 123: Wheat varieties - adoption rate

4.3 Farm Varietal Diversification and Yield Change

Results for intra-species genetic diversity and the variance of yield are reported in Figure 14. The former is analysed using the Simpson Diversity Index (SI), adjusted to account for varietal diversification of the target crops. The index ranges from zero to one, indicating respectively full specialization and full diversification. The latter is calculated by observing harvest quantity in two different points of time, for the current year (2018) and the previous one (2017).

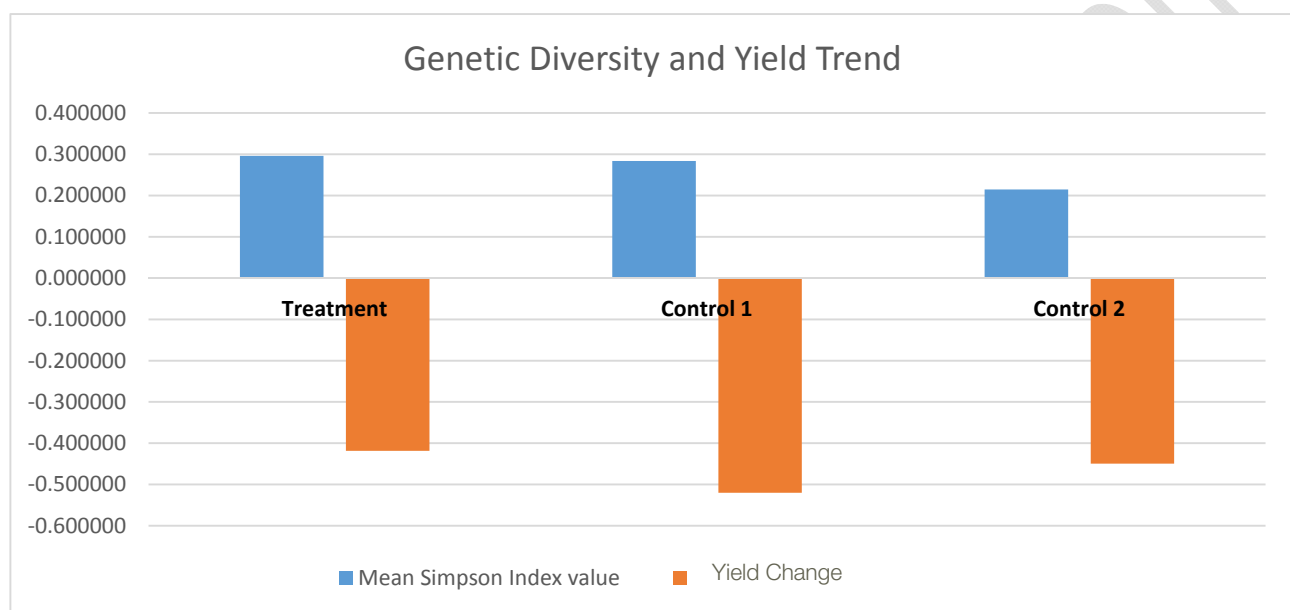


Figure 14: Descriptive results of the Simpson Index (SI) for varietal diversity and yield variation, disaggregated by sample

The treatment group appear to have a mean Simpson Index of 0.3, while the control groups 1 and 2 report a SI value of 0.28 and 0.21, respectively. This difference might be attributed to their participation in the project, and it could also explain the lowest value for control group 2, geographically too remote to be reached by a knowledge spillover effect. The yield trend is negative for both intervention and comparison groups, reflecting the distress of the agricultural sector in general. One explanation for the negative trend is likely to be found in the high exposure to climatic shocks leading to losses in rice and wheat harvests (see the section on Focus on Shock Exposure). However, farmers in the treatment group appear to have suffered less in comparison to the controls: they harvested 40% less in 2018 compared to 2017, while control groups lost 50% (control 1) and 45% (control 2) in 2018 compared to yields for 2017.

Project participants have a higher mean SI value for intra-species genetic diversity when compared to control group 2. This difference is slightly reduced for control 1, possibly indicating a spillover effect. The general trend of agricultural productivity is negative, highlighting the severe distress of the sector. However, participants appear to have witnessed a smaller rice and wheat productivity losses in comparison to control groups.

Focus on Shock Exposure

Figure 15 illustrates the proportion of respondents reporting the eight most commonly reported shocks during the reference period 2015-2017. When looking at the number of shocks reported by households we see that, across almost all events, there is a moderate difference between samples. However, the intervention group appear to have been less exposed to the decrease in sale prices of target crops and loss of harvest because of pest and disease control measures, especially if compared to non-participants from their same villages.

Interviewed households declared that the recurrence of climatic stressors leading to rice and wheat harvest loss, together with the rise in prices for agricultural inputs, mostly affected their livelihood in the past three years. Their frequency of occurrence is reported by almost 100% of households. These factors are important to contextualize the decreasing performance of rice and wheat yields in the period 2017-2018. Other shocks frequently reported are food price increases, which are affecting 80% of treatment households, 76% of control 1 households, and 75% of control 2 households, and decreases in the sale price of target crops for 69% of the treatment group, 83% of control 1 and 70% of control 2. Lastly, harvest loss due to pests and diseases affected 44% of treatment households, 67% of control 1 and 45% of control 2. Shocks related to the health status, employment conditions and destruction of productive assets are the least frequent. Agricultural-related shocks appear to have occurred with the highest frequency and regularity, reflecting the conditions of distress of the sector in general.

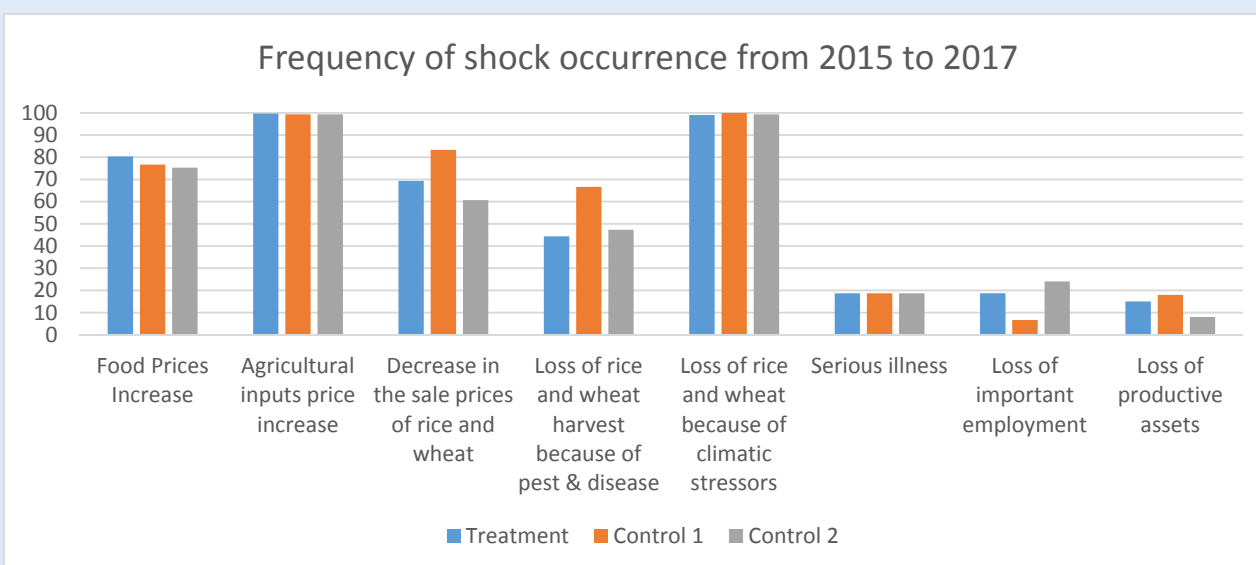


Figure 15: Frequency of shock occurrence, disaggregated by sample

4.5 Establishing Causal Relationships among Outcome Variables: Multivariate Analysis

The presentation of descriptive findings reveals that there are positive results for project participant compared with non-participants. Nonetheless, findings have so far been presented in terms of descriptive mean values, and do not provide a solid ground for establishing a causal relationship with the participation in the project. Furthermore, the consequences of a small difference in mean values might be more significant when inserted in the broader picture, accounting not only for the outcome variables but rather for the extent and direction of their relationships. Hence, multivariate statistical procedures are performed and results presented in the next sub-sections. Table 3 shows the variables used as controls and their average results across the samples.

Table 3: Outcome variables used as controls and their average results across the samples

Control variable – mean results		Treatment				Control 1				Control 2			
Variable	Description	Mean	Std. Dev	Min	Max	Mean	Std. Dev	Min	Max	Mean	Std. Dev	Min	Max
Household head gender	Dummy (1 Female)	0.25	0.4	0	1	0.3	0.46	0	1.	0.15	0.36	0	1
Household head age	Years	48.4	12	19	80	46.7	13.2	16	85	46	12.7	18	90
Household head edu	HH number of years of education	3.2	1.5	1	5	3.06	1.7	1	5	3	1.6	1	5
Household Size	Number of household members	7.7	3.2	2	20	7.23	3	2	17	7.4	3.1	2	19
Migrants	Household migrant members	0.46	0.8	0	4	0.68	1.1	0	5	0.3	0.8	0	4
Access to Information	Aggregate measure of: literacy rate, ownership of cellphones, television or radio	1.34	0.8	0	3	1.4	0.8	0	3	1.6	0.8	0	3
Subjective Poverty	Personal perception of poverty change in the reference period 2012-2017	8.22	1.3	3	10	8.5	1.5	3	10	8.3	1.6	2	10
Off-farm Income Proportion	Percentage of income gained from any off-farm activity	1.7	0.9	0	4	2	1.1	0	4	1.4	0.8	0	3
TLU	Tropical Livestock Unit	0.7	0.9	0	6.3	0.7	0.8	0	4	0.9	0.8	0	3.5
Land owned	Acres	8.1	15	0	96	7.4	14	0	108	8	13	0	49

4.6 The Three-Stage Least Square Regression Model

The three-stage least squares (3SLS) regression model is performed to observe the relationship of outputs and outcomes as shown in the Theory of Change, namely: (1) the influence of project participation on the adoption rate, (2) the varietal diversification, and the influence that diversity has on (3.1) the change of yield, on the rate of recovery from the occurrence of target crop loss because of (3.2) pest and disease, and (3.3) climatic stressors. Therefore, three different 3SLS models were run. Tables 4, 5 and 6 report their results.

Table 4: Results of three-stage least squares (3SLS) regression model 1

Model 1			Varietal Diversification			Yield Change		
Variables	Coef.	P-value	Variables	Coef.	P-value	Variables	Coef.	P-value
Participation	3.00931	0.000	Adoption	0.01982	0.005	Diversification	0.92912	0.051
HH_gender	-0.16765	0.369	HH_gender	-0.09587	0.001	HH_gender	0.14511	0.012
HH_age	0.01369	0.025	HH_age	0.00116	0.233	HH_age	-0.00258	0.086
HH_edu	0.06578	0.234	HH_edu	-0.00553	0.527	HH_edu	-0.00960	0.400
Size	0.01819	0.477	Size	-0.00379	0.347	Size	0.00273	0.615
Migrants	0.11230	0.209	Migrants	0.00224	0.873	Migrants	0.01305	0.483
Information	-0.06638	0.477	Information	0.05633	0.000	Information	-0.08497	0.004
Poverty	-0.00839	0.876	Poverty	0.00236	0.780	Poverty	-0.02962	0.008
Off-farm	-0.01897	0.794	Off-farm	0.02160	0.058	Off-farm	-0.05481	0.003
Land Owned	0.00500	0.350	Land Owned	0.00232	0.006	Land Owned	-0.00230	0.147
TLU	0.04207	0.601	TLU	0.02488	0.049	TLU	-0.06341	0.002
Constant	-0.35088	0.583	Constant	0.07001	0.484	Constant	-0.08114	0.561

Table 5: Results of three-stage least squares (3SLS) regression model 2

Model 2			Varietal Diversification			Recover from Pest&Disease		
Variables	Coef.	P-value	Variables	Coef.	P-value	Variables	Coef.	P-value
Participation	3.00931	0.000	Adoption	0.01982	0.005	Diversification	124.32360	0.028
HH_gender	-0.16765	0.369	HH_gender	-0.09587	0.001	HH_gender	9.85724	0.153
HH_age	0.01369	0.025	HH_age	0.00116	0.233	HH_age	0.11782	0.510
HH_edu	0.06578	0.234	HH_edu	-0.00553	0.527	HH_edu	1.74779	0.199
Size	0.01819	0.477	Size	-0.00379	0.347	Size	0.74067	0.252
Migrants	0.11230	0.209	Migrants	0.00224	0.873	Migrants	-3.66988	0.098
Access								
Information	-0.06638	0.477	Information	0.05633	0.000	Information	-4.24162	0.231
Poverty	-0.00839	0.876	Poverty	0.00236	0.780	Poverty	2.69384	0.042
Off-farm	-0.01897	0.794	Off-farm	0.02160	0.058	Off-farm	-1.33378	0.540
Land Owned	0.00500	0.350	Land Owned	0.00232	0.006	Land Owned	-0.37299	0.048
TLU	0.04207	0.601	TLU	0.02488	0.049	TLU	-1.29981	0.600
Constant	-0.35088	0.583	Constant	0.07001	0.484	Constant	16.27269	0.327

Table 6: Results of three-stage least squares (3SLS) regression model 3

Model 3								
Adoption			Varietal Diversification			Recover from Climatic Stressors		
Variables	Coef.	P-value	Variables	Coef.	P-value	Variables	Coef.	P-value
Participation	3.00931	0.000	Adoption	0.01982	0.005	Diversification	33.96028	0.097
HH_gender	-0.16765	0.369	HH_gender	-0.09587	0.001	HH_gender	3.40456	0.172
HH_age	0.01369	0.025	HH_age	0.00116	0.233	HH_age	-0.05289	0.413
HH_edu	0.06578	0.234	HH_edu	-0.00553	0.527	HH_edu	0.91696	0.062
Size	0.01819	0.477	Size	-0.00379	0.347	Size	0.14414	0.537
Migrants	0.11230	0.209	Migrants	0.00224	0.873	Migrants	1.29180	0.107
Information	-0.06638	0.477	Information	0.05633	0.000	Information	-3.25052	0.011
Poverty	-0.00839	0.876	Poverty	0.00236	0.780	Poverty	-0.63538	0.185
Off-farm	-0.01897	0.794	Off-farm	0.02160	0.058	Off-farm	1.18072	0.133
Land Owned	0.00500	0.350	Land Owned	0.00232	0.006	Land Owned	-0.11793	0.083
TLU	0.04207	0.601	TLU	0.02488	0.049	TLU	-2.15151	0.016
Constant	-0.35088	0.583	Constant	0.07001	0.484	Constant	22.59605	0.000

The first two regressions, which are linking participation to adoption, and adoption to varietal diversification, are the same for every model –what changes in each of them is the last dependent variable: firstly, yield change, secondly, recovery from pests and diseases, and lastly, recovery from climatic disturbances. Results demonstrate the effectiveness of the consequentiality and causality of the relationships among the selected dependent variables. While the significance of the p-value and the coefficient for the relationship participation – adoption is not surprising, the relevance of the same values for all the three outcome variables indeed shows that: (1) participation in the projects is associated to the adoption of the proposed varieties, (2) becoming an adopter has in turn influenced households to increase the genetic diversity of target crops on their farm and, lastly, (3) a more diversified rice and wheat production has positively influenced change in yield productivity and the perceived ability to recover from agricultural shocks. The findings thus confirm that an improved capacity to adapt to agriculture-related disturbances can be linked to participation in the S4N projects.

The 3SLS results confirm the hypothesis of consequential and causal relationship between participation in the project, the adoption of varieties disseminated by Bioversity and the increase in household resilience. Genetic diversity of rice and wheat is found to significantly reduce yield fluctuations and, in turn, increase the capacity to recover from target crop loss because of pests, diseases and climatic stressors.

4.7 The Doubly Robust Estimator

Once the causal relationship among project outputs and outcomes is established, a further multivariate analysis is run to estimate the impacts. For this purpose, the analysis relies on the doubly robust estimation (DR) to quantify the average causal effect of the treatment on the following outcome variables: (1) yield change, (2) varietal diversification, (3) crop diversification, (4) perceived agricultural trend, (5) recovery from a set of climatic shocks, and (6) social cohesion. Results are shown in Table 7. The treatment effect (TE) indicates the comparison between treatment and control group 2, while the Spillover Effect (SE) compares the control groups 1 and 2.

Table 7: Impact results

<i>Impact results</i>	<i>Treatment</i>	<i>Control 1</i>	<i>Control 2</i>	<i>Treatment Effect %</i>	<i>Spillover Effect %</i>
<i>Agricultural Production</i>					
Yield Change	-0.421785	-0.526453	-0.441386	-4.44	19.27
Varietal Diversification	0.305186	0.275636	0.173577	75.82	58.80
Crop Diversity	0.190149	0.267975	0.165489	14.90	61.93
Perceived agricultural trend	3.657257	3.329495	3.415456	7.08	-2.52
<i>Recovery from Agricultural Shock</i>					
Increase Food Price	40.74434	53.45142	29.58905	37.70	80.65
Increase Price Inputs	24.10070	29.78826	17.03296	41.49	74.89
Pests & Diseases	80.36785	69.03285	76.59672	4.92	-9.87
Climatic Stressors	25.26851	29.53251	15.70253	60.92	88.07
<i>Social Dynamics</i>					
Social Cohesion	5.077874	3.890228	4.348134	16.78	-10.53

The DR findings are coherent with the results demonstrated with the system of simultaneous regressions. It is possible to observe a TE on yield change inferior to almost -5%. As already indicated above, in almost all the samples there has been a reduction in rice and wheat yields, but the households belonging to the treatment group have undergone an average reduction lower than the other two control groups. The impact on the rates of recovery from agriculture-related shocks is extremely significant for all variables, particularly for climatic shocks, where the TE is equal to 60%.

The S4N Initiative in India (Bihar) has made the most significant impact in terms of rice and wheat varietal diversification (75% TE). This increased sensitivity also appears to have influenced the diversity of crops being planted, with a TE of nearly 15%. Overall, the project has also positively influenced perception on the past agricultural trend. To conclude, an indicator that is not explicitly included among the project desired impact, which has, however, provided positive results, is the effect on social cohesion, equal to 17%.

The DR findings are coherent with the results demonstrated with the system of simultaneous regressions. Significant impacts have been found in terms of yield change (-5%), perceived ability to recover from the climatic shocks (60%), and rice and wheat varietal diversification (75%). Interestingly, the project had a beneficial effect on the social cohesion of participant households (17%).

5. Conclusions

Using different quantitative methods as complementary sources to paint a picture of the ways in which the Seeds for Needs Initiative has changed lives and farm landscapes, we can draw causal links. This causal story begins with project beneficiaries adopting the project's disseminated varieties in 85% of cases, and with beneficiaries becoming generally more sensitive towards the role of intra-species genetic diversity on their farms. Greater genetic diversity of target crop cultivation has led to lower yield variance scores. Improved stability of farm production has, in turn, led to participants feeling more empowered in responding to disturbances. Farmers perceived an increase in their capacity to recover from agriculture-related shocks since Initiative implementation, as reflected in their higher recovery rate scores.

Specifically, this impact analysis has found that the S4N Initiative has had a beneficial effect on other household-level outcomes of interest, particularly a greater sense of social cohesion. This increase is likely to depend on the events organized under S4N, particularly farmers' field days and exchange visits, as they were explicitly aimed at creating a network of farmers and enabling them to establish trust and cooperative collaborations with their peers. Quantitative data also allows us to estimate a small to moderate effect in encouraging inter-species diversification of crops. We did not find any significant effect of the Initiative on food and nutrition security or in terms of increase in yield productivity.

The analysis paints a picture of an agricultural sector under extreme pressure and stress. The frequency of occurrence of agriculture-related shocks in the past three years is equal among all samples in the differing geographical areas that were visited. The fluctuation in prices of agricultural inputs, food commodities and sale price of target crops is severely affecting rural livelihoods. While frequency of crop loss due to pest and disease occurrence is moderate, the loss of harvests due to extreme climatic events appears to be extremely harsh and, paradoxically, unfluctuating. Therefore, we attribute the decreasing yield trends to climate instability (Figure 16).

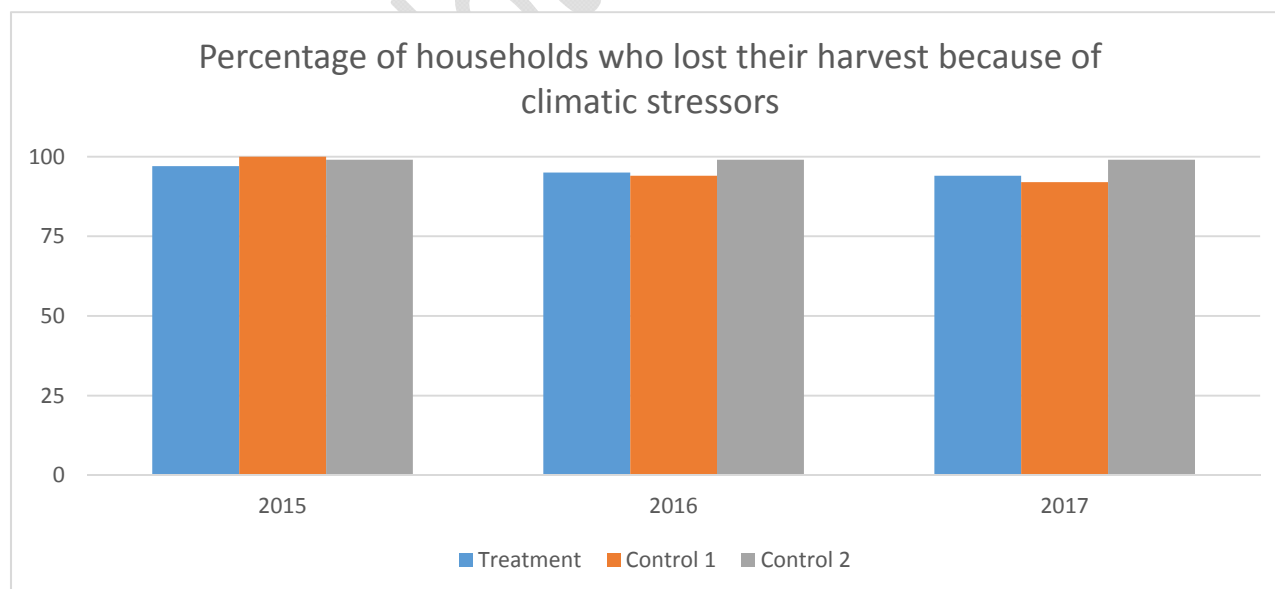


Figure16: Percentage of households losing their harvests due to climatic stressors.

Despite the challenges agricultural is facing, S4N intervention households perceive themselves to be more in control of their productivity: they score higher in the perceived recovery rate, especially when exposed to climatic shocks and fluctuations of food and input prices. They also lost less harvest compared with the other control groups. The analysis has found a strong, causal relationship between these scores and the greater varietal diversification on participants' farms.

Diversification of rice and wheat varieties was a key outcome of interest for the S4N Initiative. The Initiative logic acknowledged that agricultural diversification was becoming increasingly less viable due to decreased availability of stress-tolerant seeds. The intervention trained beneficiary households in participatory seed selection, seed conservation and multiplication practices and improved agronomic techniques. The effectiveness of these capacity building activities is proved by the spillover effect found in this study: households not participating in any of these activity were found not only to cultivate some of the S4N varieties, but also to diversify their agricultural production.

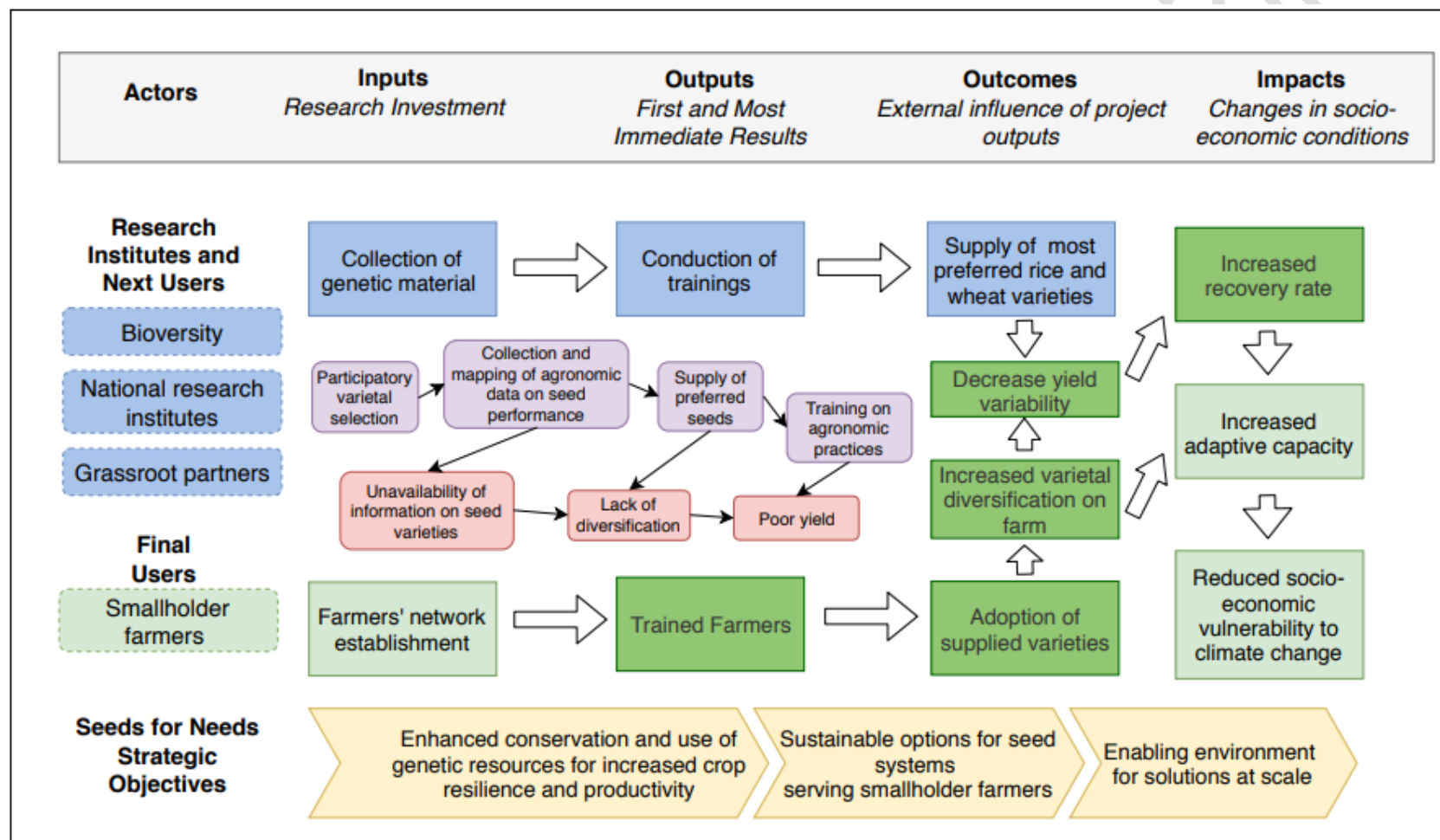


Figure 167: The S4N Initiative Theory of Change (ToC), where the dark green boxes illustrate the key areas to which the Initiative contributed beneficial impact.

6. Programme Learning Considerations

This section briefly outlines learning opportunities derived from the project that have emerged from study findings and feedback from stakeholders. The recommendations are organized along two domains: (1) program design and (2) additional research.

Programme Design

This study was conducted in the absence of any baseline data. The research team wish to stress the need to collect socio-economic data before the implementation of any further activity under the S4N umbrella. Conducting ex-ante needs assessments prior to project development is part of a robust and proper project-cycle management practice, particularly for the effective evaluation of the initial demand. It further allows activities to be tailored to the specific needs of the area of interest, and resources to be allocated more efficiently.

The analysis has found the S4N activities to have had a beneficial impact on variables related to social cohesion. However, these variables are not included in project proposals and narrative. The research team believes this is a strength of the S4N Initiative that should not be overlooked, but rather leveraged to further scale out the projects.

Additional Research

This study was conducted immediately following the conclusion of project intervention activities in Bihar, India. This has limited capturing the potential magnitude of impact that the intervention may have had, as some impact indicators take longer to take effect and be assessed. Moreover, it would be useful to understand to what extent the impact observed in this study continues to be experienced by project beneficiaries. Some key indicators to monitor in future research would include: climate variability and shock occurrence, yield performance and reach of seed dissemination.

It would be particularly useful to monitor fluctuation in climate and yields. This would ensure a more comprehensive and less perception-based measurement of the potential of the disseminated seeds, and their “insurance” potential against climate change.

References

- Adger, W. N. (2006). Vulnerability. *Global environmental change*, 16(3), 268-281.;
- Ahsan, M. N., & Warner, J. (2014). The socioeconomic vulnerability index: A pragmatic approach for assessing climate change led risks—A case study in the south-western coastal Bangladesh. *International Journal of Disaster Risk Reduction*, 8, 32-49.;
- Badmos, B. K., Adenle, A. A., Agodzo, S. K., Villamor, G. B., Asare-Kyei, D. K., Amadou, L. M., & Odai, S. N. (2018). Micro-level social vulnerability assessment towards climate change adaptation in semi-arid Ghana, West Africa. *Environment, Development and Sustainability*, 20(5), 2261-2279.;
- Caracciolo F., Furno, M. (2017). Quantile treatment effect and double robust estimators: An appraisal on the Italian Labor Market, *Journal of Economic Studies*, 44(4).
- Di Falco, S., & Chavas, J. P. (2009). On crop biodiversity, risk exposure, and food security in the highlands of Ethiopia. *American Journal of Agricultural Economics*, 91(3), 599-611.
- Di Falco, S., & Perrings, C. (2003). Crop genetic diversity, productivity and stability of agroecosystems. A theoretical and empirical investigation. *Scottish Journal of Political Economy*, 50(2), 207-216.
- Emsley, R., Lunt, M., Pickles, A., & Dunn, G. (2008). Implementing double-robust estimators of causal effects. *Stata Journal*, 8(3), 334-353.
- Funk, M. J., Westreich, D., Wiesen, C., Stürmer, T., Brookhart, M. A., & Davidian, M. (2011). Doubly robust estimation of causal effects. *American journal of epidemiology*, 173(7), 761-767.
- Gallopín, G. C. (2006). Linkages between vulnerability, resilience, and adaptive capacity. *Global environmental change*, 16(3), 293-303.;
- Gotor, E., Bellon, A., Polar, V., Caracciolo F. (2017). Assessing the benefits of Andean crop diversity on farmers' livelihood: Insights from a development programme in Bolivia and Peru, *Journal of International Development*. 29(7).
- Hahn, M. B., Riederer, A. M., & Foster, S. O. (2009). The Livelihood Vulnerability Index: A pragmatic approach to assessing risks from climate variability and change—A case study in Mozambique. *Global Environmental Change*, 19(1), 74-88.;
- Heal, G. M. (2000). *Nature and the marketplace: capturing the value of ecosystem services*. Island Press.
- Heisey, P. W., Smale, M., Byerlee, D., & Souza, E. (1997). Wheat rusts and the costs of genetic diversity in the Punjab of Pakistan. *American Journal of Agricultural Economics*, 79(3), 726-737.
- Intergovernmental Panel on Climate Change (2014) "Climate Change 2014: Impacts, Adaptation, and Vulnerability", IPCC Report;
- Intergovernmental Panel on Climate Change (2007) "Climate Change 2007: Impacts, Adaptation, and Vulnerability", IPCC Report;
- Lin, B. B. (2011). Resilience in agriculture through crop diversification: adaptive management for environmental change. *BioScience*, 61(3), 183-193.;
- Mancini, C., Kidane, Y. G., Mengistu, D. K., Pè, M. E., Fadda, C., & Dell'Acqua, M. (2017). Erratum: Joining smallholder farmers' traditional knowledge with metric traits to select better varieties of Ethiopian wheat. *Scientific Reports*, 7(1), 13076-13076.;
- Nankya, R., J. Mulumba, F. Caracciolo, M. Raimondo, F. Schiavello, E. Gotor, E. Kikulwe, and D. Jarvis, (2017). Yield Perceptions, Determinants and Adoption Impact of on Farm Varietal Mixtures for Common Bean and Banana in Uganda, *Sustainability* 9(8), 1321.
- Otto, I. M., Reckien, D., Reyer, C. P., Marcus, R., Le Masson, V., Jones, L., & Serdeczny, O. (2017). Social vulnerability to climate change: a review of concepts and evidence. *Regional environmental change*, 17(6), 1651-1662.;

- Pardey, P. G., Alston, J. M., & Ruttan, V. W. (2010). The economics of innovation and technical change in agriculture. In *Handbook of the Economics of Innovation* (Vol. 2, pp. 939-984). North-Holland.;
- Smale, M., Hartell, J., Heisey, P. W., & Senauer, B. (1998). The contribution of genetic resources and diversity to wheat production in the Punjab of Pakistan. *American Journal of Agricultural Economics*, 80(3), 482-493.
- Van Etten, J., Beza, E., Calderer, L., Van Duijvendijk, K., Fadda, C., Fantahun, B & Kiambi, D. (2016). First experiences with a novel farmer citizen science approach: crowdsourcing participatory variety selection through on-farm triadic comparisons of technologies (TRICOT). *Experimental Agriculture*, 1-22.;
- Wanjala, B. M., & Muradian, R. (2013). Can big push interventions take small-scale farmers out of poverty? Insights from the Sauri Millennium Village in Kenya. *World Development*, 45, 147-160.
- Widawsky, D., & Rozelle, S. (1998). Varietal diversity and yield variability in Chinese rice production. In *Farmers gene banks and crop breeding: Economic analyses of diversity in wheat maize and rice* (pp. 159-172). Springer, Dordrecht.
- Yachi, S., & Loreau, M. (1999). Biodiversity and ecosystem productivity in a fluctuating environment: the insurance hypothesis. *Proceedings of the National Academy of Sciences*, 96(4), 1463-1468.
- Zellner, A., & Theil, H. (1962). Three-stage least squares: simultaneous estimation of simultaneous equations. *Econometrica: Journal of the Econometric Society*, 54-78.
- Zhu, Y., Chen, H., Fan, J., Wang, Y., Li, Y., Chen, J & Mew, T. W. (2000). Genetic diversity and disease control in rice. *Nature*, 406(6797), 718.

Annex I - List of partners

- International Water Management Institute (IWMI)
- Gene Campaign (GC)
- Humana People to People India (HPPI)
- Indian Council of Agricultural Research (ICAR)
- National Bureau of Plant Genetic Resources (NBPGR)
- Nand Educational Foundation for Rural Development (NEFORD)
- Ashok Sansthan (AS)
- Action for Social Advancement (ASA)
- Deendayal Research Institute (DRI)
- Krishi Vigyan Kendra (KVK)
- Sher-e-Kashmir University of Agricultural Sciences and Technology (SKUAST)

Annex II – Consulted Documents

Utilization of Ex Situ Collections Using Climate Analogues for Enhancing Adaptive Capacity to Climate Change in South Asia	P. Mathur	Progress Report 2011-2012
Initial Establishment of a Farmer-Based Experimentation Network in the Indo-Gangetic Plains (IGP) Region: Pilot Project for On-Farm Participatory Climate Change Adaptation and Visualization	P. Mathur	Technical Report 2011
Utilization of Ex Situ Collections Using Climate Analogues for Enhancing Adaptive Capacity to Climate Change in South Asia	P. Mathur	Progress Report 2012-2013
Broadening the Genetic Base of Wheat Cultivation to Empower Farmers for Climate Change Adaptation Through Crowdsourcing	CCAFS	Newsletter, 2013
Utilization of Ex Situ Collections Using Climate Analogues for Enhancing Adaptive Capacity to Climate Change in South Asia	P. Mathur	Final Report 2013-2014
Review of Seeds for Needs projects in India	A. Singh	Consultancy Report 2017
Implementing results-based management, a multi-level approach for Bioversity International: ideas from Seeds for Needs learning lab	C. Proietti, L. Ortolani	Consultancy Report, 2017
Utilization of ex situ collections and climate analogues for enhancing adaptive capacity to climate change	S. Archak	Annual Progress Report 2012
Broadening the genetic base of crops to empower farmers for climate change adaptation through crowdsourcing	Unknown	Project Summary 2013
Utilization of ex situ collections and climate analogues for enhancing adaptive capacity to climate change	P. Aggarwal, P. Mathur, DC Bhandari	Project Proposal 2011
Use and conservation of agrobiodiversity for increased agricultural sustainability, smallholders' wellbeing and resilience to climate change in India	E. Frison	Concept Note 2012
Seeds for Needs India - Broadening the genetic base of crops to empower farmers for climate change adaptation through crowdsourcing	S. Mittra, P. Mathur, M. Dadlani, A. Gupta	Web Poster 2015
Broadening the genetic base of crops to empower farmers for climate change adaptation through crowdsourcing	G. A. Parray	Final Report 2016
Impact of Seeds for Needs intervention in Bihar	A. Gupta	Report 2016 (?)

Annex III – Disseminated Varieties

<i>Wheat varieties</i>		<i>Rice Varieties</i>	
BAZ	MPO1215	BPT5204	PUSA-1612
CBS38	PBW343	DANTESHWARI	PUSA44
CBW38	PBW502	NDR359	PUSABASMATI
CSW16	PBW550	GONTRABIDHAN1	PUSABASMATI1121
CSW18	RAJ4120	NDR97	PUSABASMATI1509
DBW14	WH147	HKR47	PUSASUGANDH4
DBW17	WH711	HUR105	PUSASUGANDH5
DBW621-50	WR544	IGKVR1	R.BHAGWATI
DPW621-50	JW1202	IGKVR2	R.MANSOORI
GW273	JW1203	IMPROVEDSAMBAMAHSURI	R.SUWASINI
GW322	JW3020	INDIRABARANIDHAN	RAJASHREE
GW366	JW3288	INDIRABARANIDHAN1	RAJENDRABHAGWATI
HD2733	K307	IR36	RAJENDRAMAHSURI
HD2824	K9107	IR64	PNR381
HD2864	LOK1	JR-201	RAJENDRASUWASINI
HD2888	MP3173	JR-503	RAJENDRASWETA
HD2894		KALANAMAK3	PRABHAT
HD2932		KOMAL-101	RANIDHAN
HD2967		KRANTI	RANJIT
HD2985		LALAT	SAHBHAGIDHAN
HI1500		MAHAMAYA	SAMBHAMASURI
HI1531		MALAVIYADHAN2	SARJU52
HI1544		MALAVIYASUGANDH4-3	PUSA1176
HI1563		Annada	SUGANDH3
HP1633		MTU1001	SUGANDH5
HUW468		MTU7029	SWARNASUB1
HW2045		NAVEEN	PRATIKSHA
JW1201		NDR-2064	WGL32100

Annex IV – Questionnaire Sections

Questionnaire Section	Content Description	Gender Disaggregation
Household Population	General information on the members of the household, their age, education and gender-disaggregated information on mobility patterns	x
Participation in Project Activities	Covering the participation in a total of 6 activities, the degree of usefulness and the extent of knowledge application	x
Farm Land Size	Land ownership and utilization	x
Rice and Wheat Cultivation	Explores the changes in target crops cultivation in the past 5 years, the role of women and men, the general impact on yield, and more detailed information on each one of the best performing varieties	x
Crop Productivity	Deatiled description of important crops cultivated in the past 12 months	x
Agricultural Inputs	Deatiled information of inputs used for cultivation and processing	
Livestock	Livestock ownership and derived products	x
Shock Exposure	Covering the occurrence frequency of a list of shocks, both agricultural-related and personal, and the extent of recovery in a 5 years timeframe	
Food Consumption	Including consumption of wild foods, food security status, and nutritional diversity	
Off-Fame Income Sources	Detailed information of other income sources, proportions and main expenses. Includes information on aid and loans	x
Poverty	Progress Out of Poverty Index (PPI), subjective measures of poverty, perceived control over future agricultural stability, trust and cooperation	

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